CashJournal — Complete Application Audit & Feature Reference (v2.0)

Updated architectural, mathematical, UI/UX, localization, and security audit of the native Android codebase through Phase 15 (v1.0.18-dev • Updated 2026-10-16 19:42 UTC+07:00).

Audit Revision v2.0 (2026-10-16 19:42 UTC+07:00): Updates the initial v1.0.14-dev audit (15:54 UTC+07:00) to cover all engineering changes across Phases 10 through 15 (v1.0.15-dev -> v1.0.18-dev). All 16 findings from Audit v1.0 (CRITICAL-1, HIGH-1..4, MED-1..5, LOW-1..6) have been verified resolved, and all 10 JUnit4 test suites (including the new PinSecurityTest) pass cleanly.

1. Chronological Audit & Engineering Time Log (UTC+07:00)

Timestamp (+07:00)	Version / Commit	Phase / Milestone	Summary of Changes & Audit Events
2026-10-15 22:00	d08e35a (v1.0.9)	Phase 8 Stabilization	Fixed onboarding permission launcher crash (fragment-ktx:1.8.1), clarified storage sandbox, refined 4-dot signal cluster.
2026-10-15 23:17	54d66d5 (v1.0.10)	Brand Identity	Integrated official CashJournal Logo v1 across adaptive launcher icons and in-app branding.
2026-10-16 12:52	91f3803 (v1.0.11-dev)	Sensory & i18n	Bilingual localization groundwork, knock/shake detector calibration, and audio chime synthesis.
2026-10-16 13:20	337ba09 (v1.0.12-dev)	Lock & Layout	Calibrated text hierarchy, layout geometry, 15s background auto-lock lifecycle, and wave mask performance.
2026-10-16 14:09	f1ce18c	Sub-Phase 9A	Room v1->v2 migration (initialBalance), zero-drift SQL derivation, backdating, full CRUD, and balance reconciliation.
2026-10-16 14:22	10b8967	Sub-Phase 9B	Versioned JSON backup export & atomic restore via Storage Access Framework (BackupManager).
2026-10-16 14:49	10c5ec1 (v1.0.13-dev)	Sub-Phase 9C	Custom payday cycle start day (1..28), expanded 12 preset categories, and manual category picker modal.
2026-10-16 15:43	28eaece (v1.0.14-dev)	Pre-Audit Baseline	Synchronized BuildConfig.VERSION_NAME display in Settings.
2026-10-16 15:54	v1.0.14-dev	Audit Report v1.0	Initial full-codebase audit completed. Identified 16 issues (1 Critical, 4 High, 5 Medium, 6 Low).
2026-10-16 16:21	7bb6203	Phase 10 Remediation	Fixed CRITICAL-1 (initialBalance currency conversion), HIGH-1 (live selectedAccount), HIGH-2 (world currency sync), HIGH-4 (cycle startingSpent), LOW-5 (New Year & DST math).
2026-10-16 16:31	5dcca0d	Phase 11 Remediation	Fixed HIGH-3 (weather Off hides widget), MED-1 (word-boundary regex \b in Category), MED-2 (buffered budget slider), MED-3 (SALARY picker & category/accent preservation).
2026-10-16 17:08	7259a0b (v1.0.15-dev)	Phase 12 Remediation	Fixed MED-4 (10,000-round salted SHA-256 PIN + PinSecurityTest), MED-5 (full ID localization), LOW-1 (18-pref backup), LOW-2 (DailyReminderScheduler 20:00), LOW-3/4/6.
2026-10-16 17:49	3edae55 (v1.0.16-dev)	Phase 13 Polish	Added isPreferencesLoaded cold-start gate (eliminating Frame-0 onboarding flash) and centered spring morph between Status Ring and Back Button.
2026-10-16 18:37	19ff083 (v1.0.17-dev)	Phase 14 Features	Added sticky immersive fullscreen, interactive PhoneSilhouettePreview in Onboarding Step 7, auto-triggered ReleaseNotesSheet, and reactive tooltip localization.

Timestamp (+07:00)	Version / Commit	Phase / Milestone	Summary of Changes & Audit Events
2026-10-16 19:31	2c112d3 (v1.0.18-dev)	Phase 15 Polish	Fixed AccountDetail back button closure, 2-row balance card + auto-scaling MaskedAmount, natural EN/ID copywriting overhaul, 1:2 phone silhouette (118x236dp), inline knock sensitivity.
2026-10-16 19:42	v1.0.18-dev	Audit Report v2.0	Post-Phase-15 re-audit completed. Verified 16/16 initial findings resolved and 10/10 JUnit test suites passing.

2. Executive Summary & Architectural Overview

CashJournal (app.cashjournal, current build v1.0.18-dev) is a native, offline-first personal finance and cash-flow application for Android (minSdk 26 through targetSdk 34). Built with Kotlin 2.0, Jetpack Compose, Room SQLite v2, Preferences DataStore, Kotlinx Serialization, OkHttp 4.12, and Phosphor Icons, the app is engineered around five non-negotiable pillars:

- 100% Free & Keyless: Zero paid libraries, zero user accounts, and zero API keys (Open-Meteo & ExchangeRate-API Open Access).
- 100% Local & Private: All financial records reside in the internal app sandbox (allowBackup="false" + XML exclusion rules).
- Zero-Drift Mathematical Integrity: Currency values stored as 64-bit integer minor units (Long); balances derived via SQL initialBalance + SUM(inflows) - SUM(outflows).
- Dual-Layer Network Lockdown: Both network_security_config.xml and HostAllowlistInterceptor restrict HTTPS traffic to 3 hosts.
- Ergonomic One-Handed UI & Sensory Feedback: Spring-animated floating SideRail, sticky immersive fullscreen, accelerometer knock/shake side-swap (ShakeDetector), 44.1kHz PCM chimes (ChimeGenerator), and GPU-rendered account blobs (AmbientField).

3. Complete Feature & Functional Reference (v1.0.18-dev)

3.1 Entry Point, Immersive Mode, App Lock & Lifecycle (MainActivity.kt)

- Cold-Start Initialization Gate: Holds a clean Slate100 surface on Frame 0 until state.isPreferencesLoaded is true, preventing any flash of OnboardingScreen before DataStore finishes reading.
- Sticky Immersive Fullscreen: Uses WindowInsetsControllerCompat.BEHAVIOR_SHOW_TRANSIENT_BARS_BY_SWIPE controlled by immersiveModeEnabled (default true) to hide system status and navigation bars.
- Daily Reminder Alarm Sync (20:00 Local Time): Reactively schedules or cancels a repeating 20:00 evening notification via DailyReminderScheduler and DailyReminderReceiver (with BOOT_COMPLETED persistence).
- Screen Privacy (FLAG_SECURE) & 15s Auto-Lock: Blocks screenshots/recording when Screen Privacy is active, and re-locks the app behind AppLockScreen after 15 seconds in the background.
- Automatic What's New Sheet: Compares prefs.lastSeenVersion != BuildConfig.VERSION_NAME on startup after onboarding to present ReleaseNotesSheet once per version upgrade.

3.2 First-Run Onboarding Flow (OnboardingScreen.kt — 8 Steps)

Step	Title	Functional Behavior & Controls
Step 1	Welcome	Displays official squircle emblem and 3 privacy guarantee tiles (ShieldCheck: No account, DeviceMobile: Stays on phone, CloudSlash: Nothing uploaded).
Step 2	Language	Instant switch between English and Bahasa Indonesia with 250ms crossfade.
Step 3	Region & Currency	Searchable Country/Currency list with vector CountryFlag icons and currency code badges (IDR, USD, EUR, GBP, SGD, JPY, AUD, MYR).
Step 4	Name	Optional 40-character name input with live SpaceGrotesk counter (N/40) and Skip action.
Step 5	Weather Privacy	Choose Approximate Location (~1 km rounded), Search City (8 popular city chips + custom query), or Not Now.
Step 6	Permissions	Interactive runtime permission cards for Coarse Location and Notifications (API 33+), plus Device Sandbox storage transparency card.
Step 7	Menu Layout	Interactive 118x236.dp (1:2 ratio) PhoneSilhouettePreview wireframe with tap-to-shake simulation and upright 0° idle rail; Left/Right SegmentedControl; Shake to Swap toggle with inline expandable Knock Sensitivity (Low/Medium/High) selector.
Step 8	Start Mode	Select Load Demo Mode (4 accounts, 10 transactions, 7.5M IDR budget) or Start from scratch, followed by CheckCircle completion screen.

3.3 Floating SideRail, Status Cluster & Shared Components

- Centered Status Ring <-> Back Button Morph (RailStatusCluster.kt): 42.dp Canvas gauge rendering a 270° live battery arc, 4 cellular signal dots (SignalStrengthHelper), Eye/EyeSlash privacy toggle, and HH:mm clock. On AccountDetailScreen, shrinks and pops via centered spring animation into a CaretLeft back button with keyed pointerInput(isNestedScreen) closure re-binding.
- Liquid Tab Drip & Reactive Localized Tooltips: 420ms stretching Cobalt blob across 5 tabs; long-press tooltips use rememberUpdatedState to reflect live EN/ID language changes immediately.
- Single-Line Auto-Scaling Wave Mask (MaskedAmount.kt): 32ms/digit wave animation into 7 dots (•••••••), plus automatic font scaling (0.82x at >=11 chars, 0.70x at >=14 chars, maxLines=1) so large balances never wrap.

3.4 Primary Screens (7 Screens) & Modal Bottom Sheets (9 Sheets)

Screen / Sheet	Key Features, Telemetry & Interactive Flows (v1.0.18-dev)
DashboardScreen	Total Net Worth hero; 28.dp Safe-to-Spend Today glass card with 64.dp circular % spent gauge and status badge (Safe, Near Limit, Over Budget, Set Budget, No Funds); Active Accounts carousel; Recent Activity (top 5); Monthly Budget card; bottom mist gradient & ScrollForMorePill.

Screen / Sheet	Key Features, Telemetry & Interactive Flows (v1.0.18-dev)
LedgerScreen	Transactions ('Riwayat Transaksi') with 4 live-count filter pills (All, Expense, Income, Transfer); instant search (notes, EN/ID categories, raw digits); chronological day grouping (TODAY, YESTERDAY with Jan 1/Dec 31 support, date headers).
AccountsScreen	World Currencies card synced with selectedWorldCurrency and CurrencyConverter.convertMinorUnits, 8 flag chips, manual refresh, and '+ World' button; 24.dp account cards with % of total pill and relative bar; Reorder Mode.
AccountDetailScreen	Live-updating single-account view; 2-row Balance Card (BALANCE label + 'Edit Balance'/'Ubah Saldo' pill on top, full-width auto-scaling MaskedAmount below); Money In (Emerald) & Money Out (Charcoal) tiles; account activity list.
AnalyticsScreen	Last 6 Months Money In / Money Out bar chart (transfers excluded); Monthly Budget calibrator with buffered Slider (commits to Room & plays chime onValueChangeFinished) and step buttons; cycle-filtered Top Spending Categories Donut Chart.
SettingsScreen	8 cards: Language (EN/ID), Main Currency, Monthly Reset Day (1..28), Security & Access (Salted SHA-256 PIN, Biometric, FLAG_SECURE, Lock Now), Sensory & Feedback (Sound, Haptics, 20:00 Daily Reminder), Menu Position (Left/Right, Full Screen Mode, Knock/Shake + Sensitivity), Weather, Data & Backup (SAF JSON Export/Restore, Replay Guide, Delete All Data, About + What's New).
AppLockScreen	Full-screen lock screen verifying 4-digit PIN via 10,000-round salted SHA-256 (or BiometricPrompt), with spring PIN dots and error shake + reject haptics.
9 Modal Sheets	LogTransactionSheet (type-aware category picker incl. Salary + SaveSplash), TransactionEditSheet (preserves manual category & expense accent), ReconcileBalanceSheet ('Edit Account Balance' / 'Difference'), NewAccountSheet, MainCurrencySheet (converts balance + initialBalance), WorldCurrenciesSheet, WeatherDetailSheet, PinSetupSheet, and ReleaseNotesSheet ('What's New in v1.0.18-dev').

4. Data Architecture, Security & Automated Test Suite

4.1 Zero-Drift SQL Ledger, DST-Safe Cycle Math & Salted PIN Security

```
balance = initialBalance + SUM(INCOME where src=id) + SUM(TRANSFER where dst=id) - SUM(EXPENSE where src=id) - SUM(TRANSFER where src=id)
```

- Main Currency Conversion Integrity: Converts both balance and initialBalance across all accounts, converts all transactions and monthly plan amounts, and executes recalculateAllAccountBalances() inside an atomic Room transaction.
- DST-Safe & Cycle-Aware Safe-to-Spend: Uses Math.round(diffMillis / 86_400_000.0) for remaining days and matches plan.monthKey against cycleMonthKey so startingSpent persists across calendar month boundaries during custom cycles (days 2..28).
- 10,000-Round Salted SHA-256 PIN Storage: Generates a 16-byte SecureRandom salt (saltHex:hashHex), iterates 10,000 SHA-256 rounds, and verifies in constant time via MessageDigest.isEqual() with legacy 64-hex fallback.
- Full 18-Preference JSON Backup (BackupManager.kt): Exports and restores all 18 non-sensitive preferences alongside accounts, transactions, and monthly plans.

4.2 Automated Unit Test Suite Status (10 Test Suites — 100% PASS)

Test Suite	Verified Assertions & Coverage	Result
AcceptanceCheckTest	Build Brief Section 16 acceptance numbers (Rp 18.45M net worth, Rp 150.294/day on Oct 14, 66% ring, Rp 1M expense -> Rp 91.470/day & 79% Near Limit, transfer isolation, knock sensitivity, 7-dot mask).	PASS
LedgerIntegrityTest	Zero-drift derivation on retroactive edits/deletes, atomic transfer conservation, manual balance reconciliation delta math, backdating, and initialBalance currency conversion zero-drift.	PASS
SafeToSpendTest	Leap-year February handling, status thresholds, last-day clamping, custom cycleStartDay (1, 25, boundary clamping 0 & 31), cross-month cycle expense filtering, and cross-month startingSpent persistence.	PASS
BackupRestoreTest	BackupPayload JSON round-trip serialization (all 18 PreferencesBackupDto fields), corrupted JSON failure handling, version <= 0 rejection, and zero-drift recalculation after restore.	PASS
PinSecurityTest	New in Phase 12: Verifies saltHex:hashHex format (32:64 hex chars), per-call salt randomness, 10,000-round salted verification, legacy 64-hex unsalted fallback, and malformed hash safety.	PASS
CategoryDetectorTest	Keyword detection across categories plus word-boundary negative tests confirming 'theater'/'sweater', 'chair'/'repair', and 'cost' do not trigger false positives.	PASS
CurrencyConverterTest	Same-currency identity, USD <-> IDR decimal scaling (2 vs 0 decimals), EUR -> JPY cross-rates, and missing-rate fallback.	PASS
HostAllowlistTest	Verifies HostAllowlistInterceptor permits the 3 allowed hosts and throws SecurityException on unauthorized domains.	PASS
SignalStrengthHelperTest	Monotonic translation of 4-bar and 5-bar cellular signal scales into 4 UI dots.	PASS
WeatherIconMapperTest	WMO weather code mapping across day/night variants to Phosphor icons.	PASS

5. Audit Findings Resolution Ledger & Residual Observations

5.1 Resolution Ledger of Initial Audit v1.0 Findings (16 / 16 Resolved)

ID	Severity	Initial Finding Summary	Resolution Status & Verification
CRITICAL-1	Critical	changeMainCurrency() omitted acc.initialBalance conversion.	RESOLVED (Phase 10) — Converts initialBalance + recalculates all balances; verified in LedgerIntegrityTest.

ID	Severity	Initial Finding Summary	Resolution Status & Verification
HIGH-1	High	Stale selectedAccount snapshot on AccountDetailScreen.	RESOLVED (Phase 10) — Dynamically resolved from rawAccounts in MainViewModel.uiState.
HIGH-2	High	AccountsScreen ignored selectedWorldCurrency & used IDR fallback.	RESOLVED (Phase 10) — Synced with DataStore selectedWorldCurrency & CurrencyConverter.
HIGH-3	High	Weather 'Off' (not_now) still showed fake 24°C Sun in GreetingBar.	RESOLVED (Phase 11) — Hides widget when Off; shows Thermometer + '--' when pending.
HIGH-4	High	startingSpent dropped to 0 mid-cycle when cycleStartDay > 1.	RESOLVED (Phase 10) — Matches plan.monthKey against cycleMonthKey; verified in SafeToSpendTest.
MED-1	Medium	Substring false positives in Category.detectFromNote.	RESOLVED (Phase 11) — Pre-compiled word-boundary regex (\b); verified in CategoryDetectorTest.
MED-2	Medium	Budget Slider spammed DB writes & chimes on every drag frame.	RESOLVED (Phase 11) — Buffered locally in sliderBudget; commits onValueChangeFinished.
MED-3	Medium	SALARY hidden in LogTransactionSheet; manual category/accent overwritten.	RESOLVED (Phase 11) — Type-aware category list, manual category preservation, accent retention.
MED-4	Medium	Unsalted SHA-256 PIN hashing in UserPreferencesRepository.	RESOLVED (Phase 12) — 10,000-round salted SHA-256 + SecureRandom; verified in PinSecurityTest.
MED-5	Medium	Incomplete Bahasa Indonesia ('in') localization across UI/toasts.	RESOLVED (Phases 12-15) — Full bilingual copywriting & reactive side-rail tooltips.
LOW-1..6	Low	Backup prefs, Daily Reminder scheduler, haptics/chime gating, DST/New Year dates, dead code.	RESOLVED (Phases 10-12) — 18-pref backup, 20:00 AlarmManager scheduler, haptics checks, DST rounding.

5.2 Residual Minor Observations in v1.0.18-dev (Low Priority)

- NEW-LOW-1 (AndroidManifest.xml:33-40): DailyReminderReceiver is declared with android:exported="true" and an unguarded custom intent action (app.cashjournal.ACTION_DAILY_REMINDER). Because DailyReminderScheduler uses an explicit component intent, setting android:exported="false" prevents third-party apps from spoofing reminder broadcasts while still receiving BOOT_COMPLETED and AlarmManager intents.
- NEW-LOW-2 (MainAppScaffold.kt:480 & MainCurrencySheet.kt): Selecting a world currency in MainAppScaffold.kt:480 calls showToast("Currency changed to $code") instead of strings.currencyChangedToast.format(code), and MainCurrencySheet.kt dialog strings remain in English.
- NEW-LOW-3 (app/build.gradle.kts:41-48): Release build type uses sharedDebug keystore and isMinifyEnabled = false for seamless local APK upgrades; switch to a production release keystore and enable R8 before Play Store distribution.
- NEW-LOW-4 (ui/preview/*): Historical Phase 2–3 preview composables (FoundationPreviewScreen.kt, Phase3ComponentsScreen.kt) remain unreferenced in main source sets.